

Marshall B. Frye

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SUMMARY

Materials scientist (PhD, May 2026) with 7+ years of hands-on experience in structural characterization, microstructural investigation, and process-property analysis across oxide, metal, and chalcogenide systems. Skilled in SEM/EDS, XRD, XRR, XPS, AFM, and quantitative image analysis, with a strong track record of diagnosing defect mechanisms and linking microstructure to performance. Experienced producing clear technical reports and communicating findings to multidisciplinary teams and external stakeholders.

EDUCATION

Georgia Institute of Technology

May 2026

PhD, Materials Science and Engineering

- Presidential Fellow (2021–2025). Advisor: Prof. Lauren Garten.
- Awards: MRS Student Award (2026), GT MRS Poster Session 1st place (2026), AVS Russell & Sigurd Varian Award (2025), Acta Student Award (2025), AVS Thin Film Graduate Award (2025), Novelis Graduate Scholar.

University of Florida

May 2021

B.S., Materials Science and Engineering | Minor, Business Administration

- Summa Cum Laude.

TECHNICAL SKILLS

Characterization: XRD, XRR, XPS, SEM/EDS, AFM, Raman spectroscopy, ellipsometry, SIMS, electrical impedance spectroscopy, piezoresponse force microscopy (PFM), UV-Vis spectroscopy, photoluminescence (PL), four-point probe resistivity mapping; metallographic sample preparation, hardness testing, quantitative image analysis; surface and interface characterization of coating and structural integrity.

Thin-Film Deposition & Coatings: Pulsed laser deposition (PLD), molecular beam epitaxy (MBE), electron-beam evaporation, sputter deposition, spin-coating, ALD-compatible process design; wafer-scale process development, scale-up, and process window evaluation; dielectric, ferroelectric, and metallic coating architectures.

Materials Systems: Dielectric and ferroelectric oxides ($\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$, Al_2O_3 , rare-earth ferrites), 2D chalcogenides (SnSe), platinum and metallic interlayers, oxide–metal and oxide–oxide heterostructures, interface engineering.

Experimental Design & Analysis: Design of experiments (DoE), statistical analysis of process–property data, spatial uniformity and yield mapping, phase-field modeling, machine learning (WEKA).

Software & Tools: Python, MATLAB, Microsoft Office 365, LLM tools.

Languages: English (native), French (conversational), Spanish (conversational).

RESEARCH EXPERIENCE

Graduate Research Assistant — Garten Research Group

Aug 2021–May 2026

Georgia Institute of Technology, Atlanta, GA

- Characterized 300+ thin-film specimens across oxide, chalcogenide, and metal systems, applying SEM/EDS, XRD, XPS, AFM, and four-point probe mapping to investigate microstructure, composition, and defects.
- Engineered ferroelectric–dielectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{Al}_2\text{O}_3$ heterostructures, designing novel architectures that tripled the coercive field through interfacial charge engineering work, relevant to dielectric coating design for capacitor performance optimization.
- Developed wafer-scale deposition of 2D SnSe with monolayer-thickness precision, defining characterization protocols to assess uniformity and identify structural anomalies across the full wafer.
- Applied DoE and statistical analysis to systematically diagnose root causes of inconsistent film performance, identifying deposition parameter windows that improved crystallinity, adhesion, and spatial yield across multiple material systems.
- Conducted surface and interface characterization (XPS, AFM, XRR, SEM/EDS) to evaluate film composition and structural integrity, diagnosing defect and failure mechanisms and translating findings into process corrections, producing written technical summaries communicated to collaborators and external stakeholders.
- Led competitive proposals and secured \$150,000 in user-facility access (NSF PARADIM, 2DCC), demonstrating grant writing, project leadership, and the ability to drive multi-institutional research.

- Managed procurement, refurbishment, and commissioning of \$100K+ in laboratory deposition and characterization equipment to establish a new research lab.
- Mentored five undergraduate researchers and managed multiple parallel research projects from concept through to characterization, consistently meeting milestones.

Undergraduate Research Assistant — Nino Research Group

Aug 2018–May 2021

University of Florida, Gainesville, FL

- Optimized solution-based synthesis of reticulated ceramic foams for solar thermochemical hydrogen production, gaining hands-on experience with scalable solution-processing methods for inorganic materials.
- Developed a novel sulfonation method for nanocellulose to increase functional group density, enabling enhanced surface adsorption performance.
- Led research on functionalized nanocellulose–chitosan composite filters for opioid simulant removal from water, resulting in a first-author peer-reviewed publication.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Early Career Network Representative

Sep 2024–Oct 2025

DOE Basic Energy Sciences — 3D Ferroelectric Microelectronics EFRC

- Served as liaison between a multi-institutional DOE Energy Frontier Research Center and the broader BES community; organized cross-center webinars and workshops.

Treasurer, Materials Research Society — Georgia Tech Chapter

Aug 2024–Dec 2025

- Managed chapter budget and fundraised \$5,000+ for an annual poster symposium; planned 6+ professional events annually.

Teaching Assistant

Jan 2022–Dec 2024

Georgia Institute of Technology

- Taught and supported 100+ students across Electronic/Optical/Magnetic Properties and Principles of Engineering Materials courses.

SELECTED PUBLICATIONS

- M.B. Frye et al., “Tuning the Coercive Field in Ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2\text{-Al}_2\text{O}_3$ Heterostructures via Interfacial Charge Dynamics” *Adv. Mater.* *Submitted*
- M.B. Frye et al., “Wafer-Scale Deposition of 2D SnSe with Monolayer-Thickness Precision” *Nano Letters* *In Peer Review*.
- M.B. Frye et al., “Stabilizing Metastable Rare-Earth Ferrites on (111) Platinum via an Iron Oxide Interlayer” *ACS Appl. Electron. Mater.* (2026)
- M.B. Frye et al., “Thickness-Modulated Crystal Structure and Band Gap of 2D SnSe Deposited by MBE,” *NPJ 2D Mater. Appl.* (2026).
- M.B. Frye et al., “Interlayer-Mediated Stabilization of Metastable $\text{P6}_3\text{cm ScFeO}_3$ on Al_2O_3 ,” *Adv. Mater. Interfaces* (2025).
- M.B. Frye et al., “Enhancing the Conductivity and Crystallinity of (111) Platinum Films,” *Scr. Mater.* 256, 116442 (2025).
- M.B. Frye & L. Garten, “Reaching the Potential of Ferroelectric Photovoltaics,” *Acc. Mater. Res.* 4 (2023).
- M.B. Frye et al., “Effect of Degree of Sulfonation in Nanocellulose/Chitosan Composite on Adsorption of Cationic Dye,” *Cellul. Chem. Technol.* 55, 87–99 (2021).

8 first-author publications | 14+ total publications | 10+ conference presentations